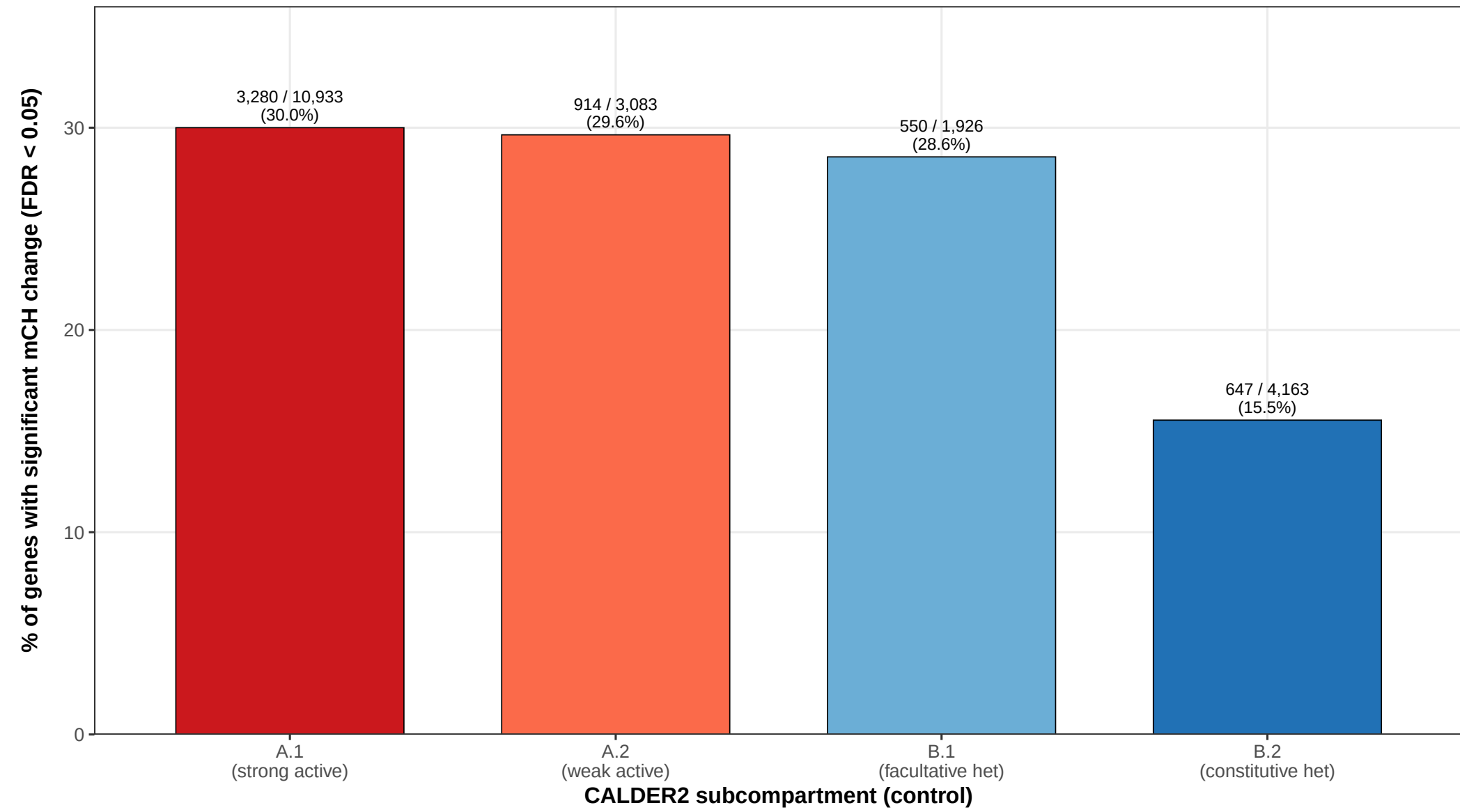


# mCH change stratified by CALDER2 subcompartment

20,105 genes assigned to 23,874 labelled 100 kb bins

## Significant mCH genes by CALDER2 subcompartment

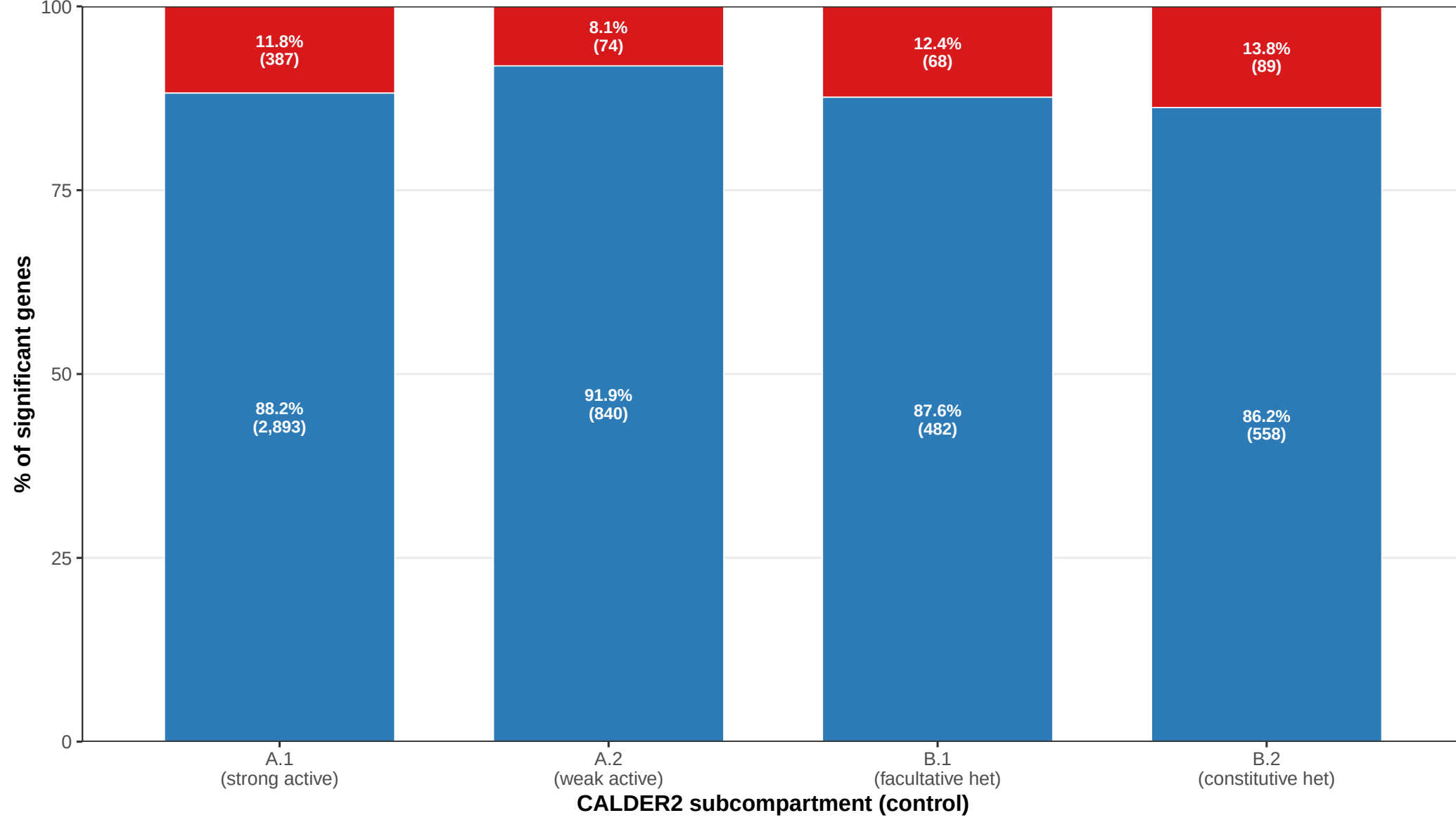
Chi-squared subcompartment x significance:  $p < 2.2e-16$



## mCH change direction by subcompartment

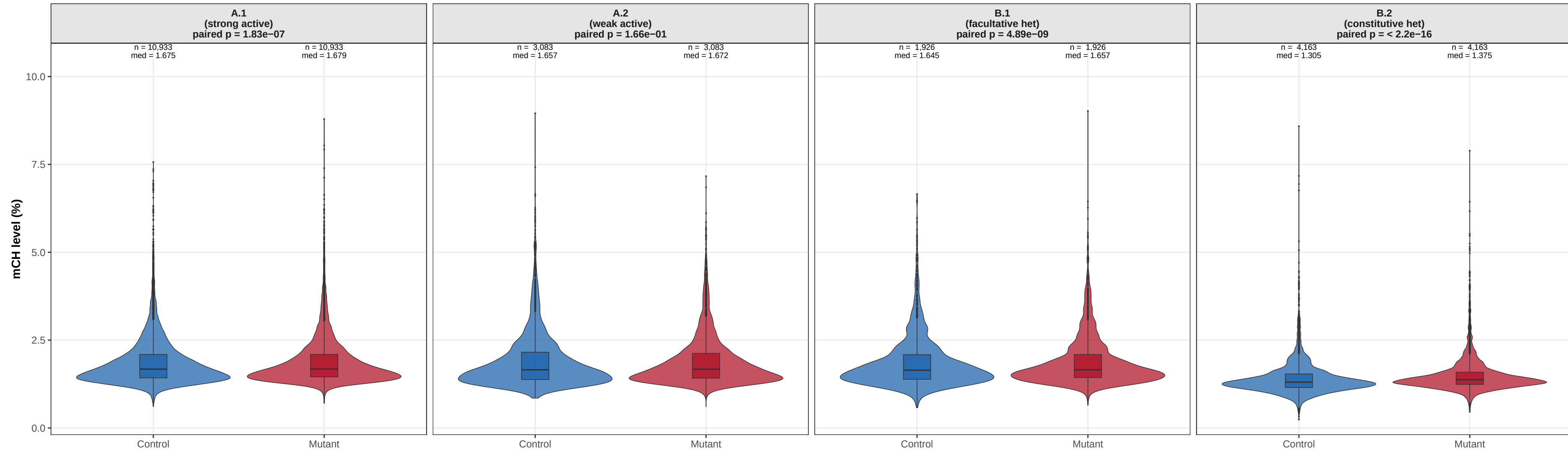
Significant genes only | chi-squared subcompartment x direction:  $p = 2.45e-03$

Direction ■ Hypermethylated ■ Hypomethylated



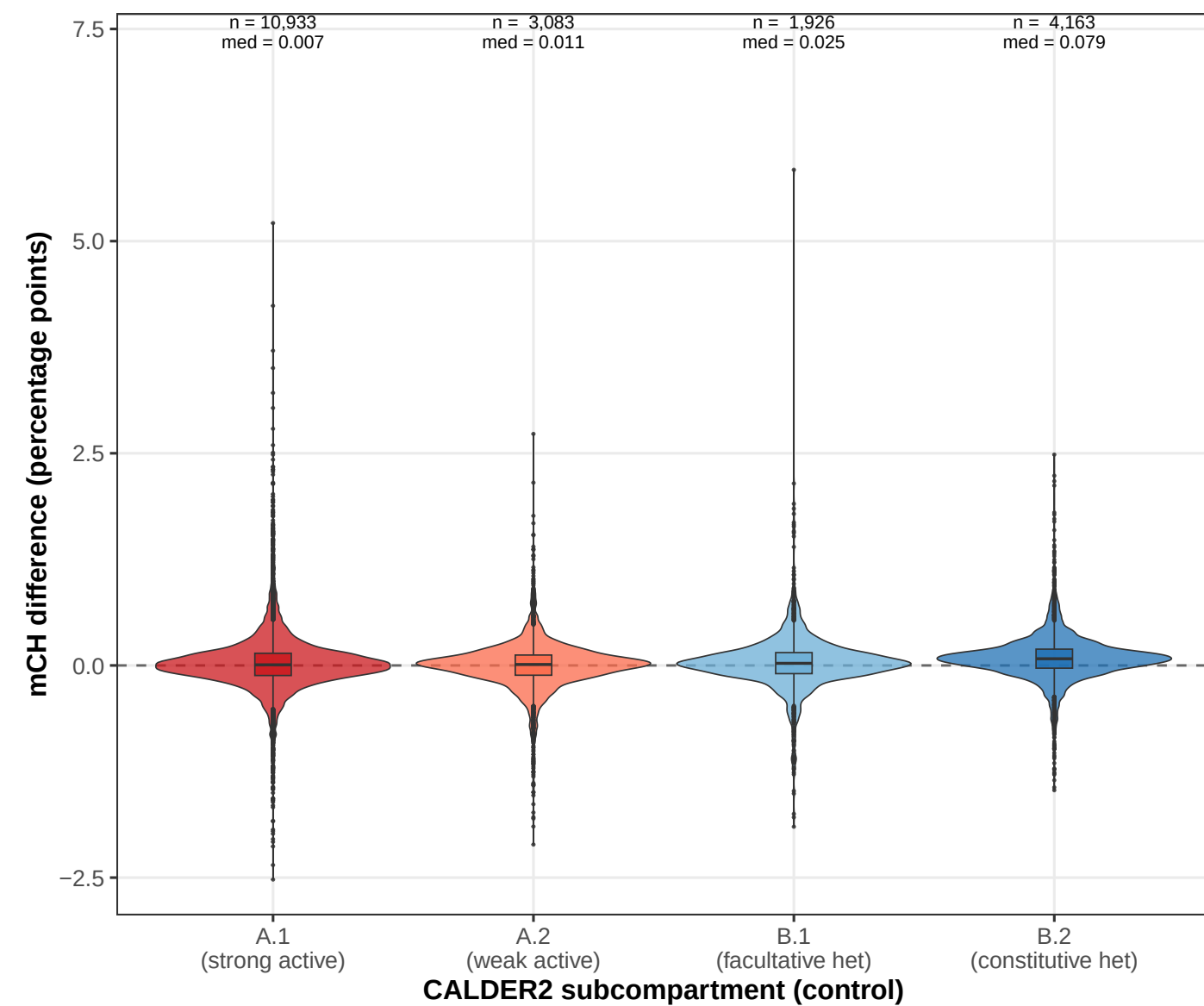
## Gene-body mCH level by subcompartment

Kruskal-Wallis of control mCH across subcompartments:  $p < 2.2e-16$



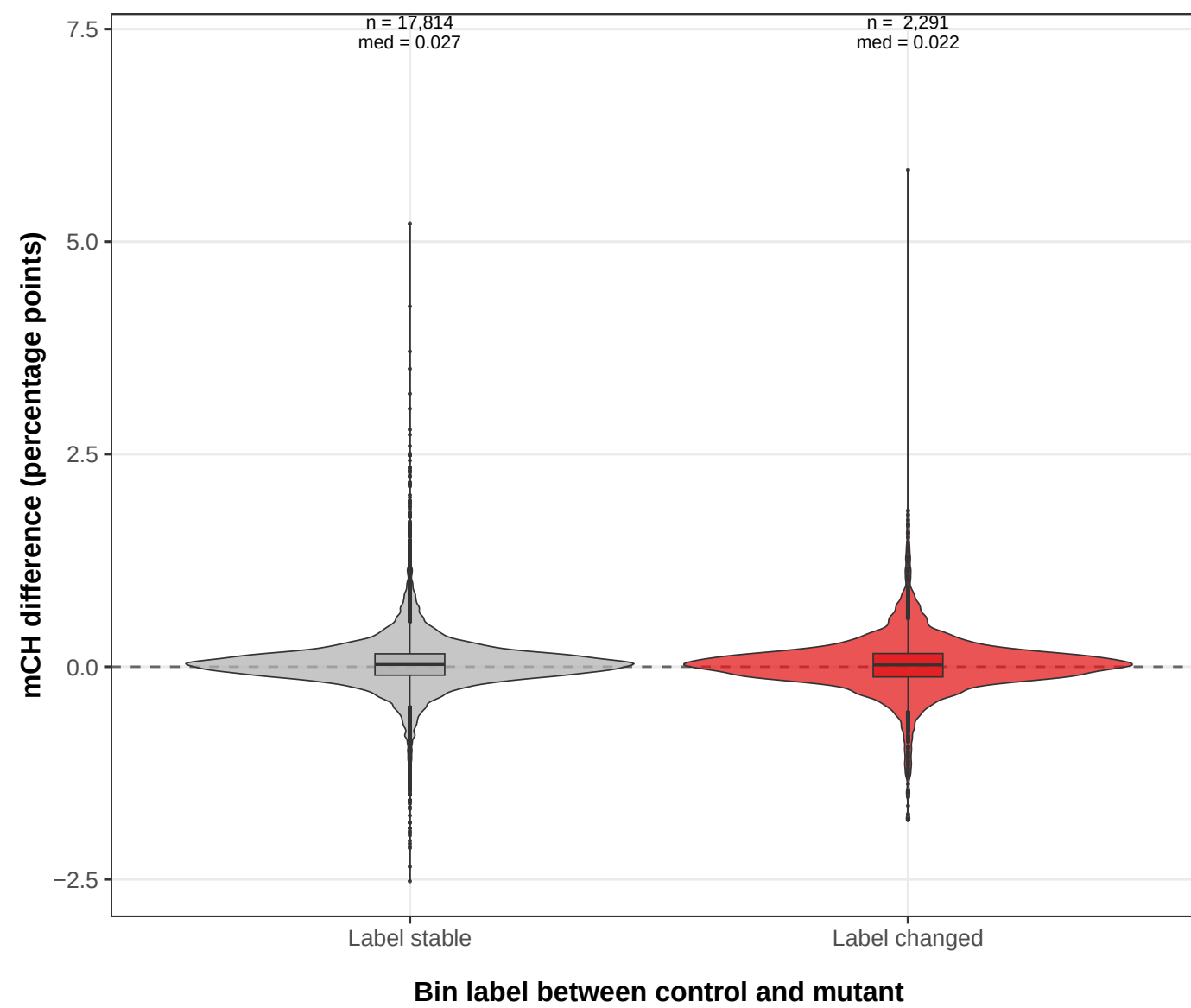
## mCH difference by subcompartment

Mutant minus control | Kruskal-Wallis across subcompartments:  $p < 2.2e-16$



## mCH difference by subcompartment label change

Wilcoxon rank-sum:  $p = 1.80e-01$



## Histone mark overlap by subcompartment

Fraction of gene bodies overlapping a consensus peak

Consensus peak ■ H3K27me3 ■ H3K27ac

